Daniele **De Martini** PhD, MIEEE

Associate Professor in Mobile Robotics @ Oxford Robotics Institute & Oxford e-Research Centre

daniele@robots.ox.ac.uk • daniele-de-martini • danieledema • Scholar • ori-mrg.github.io

Research Interests

Research Vision I pioneer **site-driven robotics**. My work departs from traditional robot-centric autonomy by treating the *site* as the primary actor and emphasising infrastructure–robot co-design. I leverage existing site infrastructure (e.g., CCTV, communications, edge/cloud compute), combined with minimal targeted augmentation (such as sparse markers), to enable reliable operation of simple robots in legacy, human-shared environments—without full site redesign. This *Site-as-the-Actor* approach bridges perception, cyber–physical systems, and human–robot interaction, enabling rapid, low-cost deployment of autonomous systems in critical infrastructure, including warehousing, healthcare, and energy.

Research Interests Multimodal perception leveraging site infrastructure (CCTV, edge compute, communications, etc) • Robot-infrastructure interplay and co-design for safety-critical shared spaces • Long-term autonomy and adaptation in dynamic human environments.

Application interests Autonomous logistics & warehouse automation (intra-facility navigation, pick-and-place) • Healthcare robotics (hospital navigation, sterile environments, human safety) • Energy infrastructure (inspection, monitoring, maintenance in constrained spaces)

Employment

University of Oxford - Oxford Robotics Institute & Oxford e-Research Centre — Oxford, UK
Associate Professor in Mobile Robotics — 10/2025 – Ongoing

University of Oxford - Keble College — Oxford, UK
Tutorial Fellow — 10/2025 – Ongoing

University of Oxford - Oxford Robotics Institute — Oxford, UK
Departmental Lecturer in Mobile Robotics — 10/2023 – 10/2025

University of Oxford - Pembroke College — Oxford, UK
Part-time Stipendiary College Lecturer — 09/2022 – 10/2025

Frontier Development Lab — Virtual
Faculty in the Virgil2.0 Team — 06/2024 – 08/2024

- Development of algorithms for providing advance warnings of hazardous solar eruptions

University of Oxford - Oxford Robotics Institute — Oxford, UK
PostDoctoral Research Assistant — 05/2018 – 10/2023

- Developed algorithms for robust odometry/localisation using radar
- Applied ML for environment understanding
- Explored robot–infrastructure interplay

University of Oxford - Kellogg College — Oxford, UK
Junior Research Fellow — 10/2019 – 09/2022

Frontier Development Lab — Virtual
Researcher in the Machine-Learning-Payloads Team — 06/2021 – 08/2021

- Development of algorithms for unsupervised change detection on satellite data through Machine Learning techniques for downlink prioritisation

Frontier Development Lab — Virtual
Researcher in the Digital-Twin-Earth Team — 06/2020 – 08/2020

- Development of algorithms for precipitation forecasts through Machine Learning techniques

FCA — Torino, Italy
Engineer Specialist in the R&D chassis group — 04/2014 – 01/2015

- Multibody design and analysis of suspensions
- Constrained optimization of suspension springs

Education

Università degli Studi di Pavia — Pavia, Italy
Ph.D. student — 02/2015 – 01/2018 (awarded Dec 2018)

- Supervisor: Tullio Facchinetti
- Thesis: *Actuator Management of a Quadcopter with Double-Axis Tilting Rotors*

University of New Brunswick — Fredericton, Canada
Visiting Ph.D. student — 10/2016 – 03/2017

- Supervisor: Howard Li

Politecnico di Torino — Torino, Italy
M.Sc. in Mechatronics Engineering — 09/2011 – 12/2013

- Supervisor: Giancarlo Genta
- Collaboration with Mediamotive
- Thesis: *Design of a VTOL UAV* (final mark: 110/110)

Università degli Studi di Pavia — Pavia, Italy
B.Sc. in Mechanical Engineering — 09/2008 – 09/2011

- Supervisor: Carlo Rottenbacher
- Thesis: *200 m K1 kayak race simulation: cinematic and dynamic analysis and physiological aspects* (final mark: 110/110)

Grants, Funding and Awards

Funding leadership: £440k as PI (independently led grants). Funding experience: £6.6M as co-I/contributing writer, including theme/work-package leadership roles within larger programmes.

EPSRC IAA – Robotic inversion – forklift operation proxy — 2026
PI and main writer — 76k £

NVIDIA Academic Grant Program – Generative AI and adversarial search to prevent robot failures — 2026
PI and main writer — Equipment ∼40k £

Keble College Research Support Scheme – Data-sharing infrastructure — 2026
PI and main writer — 3k £

Oxford UCSF – Inverted Automation: hyper-low-cost, scalable warehouse automation — 2026
PI and main writer — 96k £

InnovateUK – A general-purpose configuration management system validated in automated logistics — 2025
PI and main writer, led by Omniscient ltd — 130k £ (total 500k £)

EPSRC IAA – Robotic Inversion for Smart Logistics — 2024 – 2025
PI and main writer — 86.5k £

EPSRC IAA – RobotCycle, Mapping and Understanding Road Infrastructure and Road User Behaviour — 2024
Main writer, Co-I, additional funds — 24k £

EPSRC IAA – Autonomous Inspection for Environmental Impact and Sustainability Monitoring — 2022 – 2023
Contributing writer, named researcher — 240k £

EPSRC PG – From sensing to collaboration — 2021 – 2026
Contributing writer for the Navigation theme and the Agriculture flagship — 1M £ (6M £ total, 6 PIs)

Assuring Autonomy – Sense-Assess-eXplain — 2019 – 2021
Contributing writer and Researcher Co-Investigator — 314k £

Visiting Student – University of New Brunswick — 2016 – 2017
Visiting PhD student, Fredericton, Canada — 3k €

Ph.D. Italian Scholarship – University of Pavia — 2015 – 2018
Funded PhD school of Engineering — 36k €

Teaching Experience

Graduate departmental teaching

University of Oxford, Centre of Doctoral Studies AIMS, robotics week orgniser — 2025 – Ongoing
University of Oxford, MSc in Autonomous Robotics, Robotics Group project — Launching 2026
University of Oxford, MSc in Autonomous Robotics, Autonomous Robotics Systems — Launching 2026

Undergraduate departmental teaching

University of Oxford, *B3 – third-year group project* on *System and Software Design of a Multiagent Aerial Robotic System* — 2023 – Ongoing
University of Oxford, *B16 – software engineering* laboratory organiser — 2023 – Ongoing

Undergraduate college teaching

University of Oxford, *P1 – Mathematics* tutorials @ Keble College — 2025 – Ongoing
University of Oxford, *A2 – Electronic and Information Engineering* tutorials @ Keble College — 2025 – Ongoing
University of Oxford, *A3 – Structures, Materials and Dynamics* tutorials @ Keble College — 2025 – Ongoing
University of Oxford, *P3 – Structures and Mechanics* tutorials @ Pembroke College — 2022 – 2025

Teaching Assistant

University of Oxford, Lead Tutor for *B16 – software engineering* tutorials — 2023 – 2025
University of Oxford, *B16 – software engineering* tutorials — 2021 – 2025

Laboratory Supervision

University of Oxford, *B16 – software engineering* laboratory demonstrations — 2018 – 2023
University of Oxford, *P5 – computing* laboratory demonstrations — 2018 – 2022
University of Oxford, *Robotics* Laboratory demonstrations — 2018 – 2019
Università degli Studi di Pavia, C programming fundamentals — 2015 – 2017
Università degli Studi di Pavia, lab supervision in Prof. Facchinetti's *Robotics* course — 2015 – 2017

Invited Lectureship

Aalborg University (DK), invited lecture on *robot sensors* in Prof. Marchegiani's *Robot Navigation* course — 2021
Università degli Studi di Parma (IT), invited lecture on *satellite imagery as a robotic sensor* in Prof. Marchegiani's *3D Perception and Learning-based Data Fusion* course — 2023

Other

University of Oxford, mentoring for MSc and PhD students @ Keble College — 2025 – Ongoing
University of Oxford, mentoring for MSc and PhD students @ Pembroke College — 2024 – 2025
University of Oxford, mentoring for MSc and PhD students @ Kellogg College — 2019 – 2022
University of Oxford, Associate Fellow of the Higher Education Academy — 2021
University of Oxford, PAT (Physics Aptitude Test) marker — 2020
Università degli Studi di Pavia, Co-supervision of B.Sc. and M.Sc. students thesis — 2016 – 2017

© Licensed Patents and Intellectual Properties

IP co-inventor : *Precise Ego-Motion Estimation and Localisation for Automotive Scanning Radars* — 2018

IP co-inventor : *NeRFloors: Conditional street-level scene generation from Bird's Eye View semantic maps via Neural Radiance Fields* — 2023

Patent co-inventor : *NeRFloors: Conditional street-level scene generation from BEV (Bird's Eye View) semantic maps via NeRF (Neural Radiance Fields)* — 2023

Scientific Leadership & Community Service

Editorial Role: Associate Editor @ *Robotics and Automation Letters* (RAL) (2024 –) • Guest Editor @ Special Issue on *Trust through eXplAInability (XAI), Robustness and Verification of Autonomous Systems @ IET Radar, Sonar & Navigation* (2024) • Associate Editor @ *Intelligent Vehicles Symposium* (IV) (2024 – 2025) • Associate Editor @ *International Conference on Intelligent Transportation Systems* (ITSC) (2019 – 2024) • Associate Editor @ *International Conference on Robotics and Automation* (ICRA) (2021 – 2022)

Chair: *International Conference on Intelligent Robots and Systems* (IROS) (2022, 2025) • *International Conference on Robotics and Automation* (ICRA) (2022, 2024) • *International Conference on Intelligent Transportation Systems* (ITSC) (2019)

Organising committee: Workshop on *Quantum Robotics Standardisation: Challenges and Roadmap @ European Robotics Forum* (ERF) (2027) • Workshop on *The More the Better: Multimodal Perception for Safer Autonomous*

Navigation @ *Conference on Automation Science and Engineering* (CASE) (2025) • Industry panel on *Future Telecommunication Infrastructure and Robotics for Safety-Critical Industries and Services: Industry Demands, Safety Requirements, Technical Challenges & Solution, and Economics* @ *International Conference on Communications* (ICC) (2025) • Tutorial on *Emerging Technologies, 5G and Beyond: Task-oriented Co-design of Sensing, Communication, and Control for Cyber-Physical Systems* @ *Vehicular Technology Conference* (VTC) (2025) • Special session on *SS26: Human-Centred Intelligent Systems* @ *Iternational Conference on Industrial Technology* (ICIT) (2024) • Workshop on *Beyond Traditional Sensing for Intelligent Transportation* @ *International Conference on Intelligent Transportation Systems* (ITSC) (2023) • Special session on *Beyond Traditional Sensing for Intelligent Transportation* @ *International Conference on Intelligent Transportation Systems* (ITSC) (2019 – 2022) • Tutorial session on *Self-Supervision on Wheels* @ *European Conference on Computer Vision* (ECCV) (2022)

Program committee: *Association for the Advancement of Artificial Intelligence* (AAAI) (2024-2025) • *International Conference on Emerging Technologies and Factory Automation* (ETFA) (2019 – 2023) • *International Conference on Agents and Artificial Intelligence* (ICAART) (2019 – 2020) • *International Joint Conference on Artificial Intelligence – Pacific Rim International Conference on Artificial Intelligence* (IJCAI-PRICAI) (2020) • *European Conference on Artificial Intelligence* (ECAI) (2020)

Other reviewing activity: *Robotics and Automation Letters (RA-L)* • *Autonomous Robots (Auro)* • *Transaction on Robotics (T-RO)* • *Intelligent Vehicles Symposium* (IV) • *International Conference on Intelligent Transportation Systems* (ITSC) • *International Conference on Intelligent Robots and Systems* (IROS) • *IEEE Transactions on Mechatronics* • *Nature Machine Intelligence* • *MDPI Actuators* • *Manning Publications* • *International Symposium on Experimental Robotics* (ISER) • *International Conference on Industrial Cyber-Physical Systems* (ICPS) • *International Conference on Industrial Electronics for Sustainable Energy Systems* (IESES) • *International Conference on Cyber-Physical Systems* (ICCPS) • *World Scientific*

Memberships IEEE Membership (since 2018) • IEEE Intelligent Transportation Systems Society Membership (since 2019) • IEEE Robotics and Automation Society (since 2022) • UK Research and Innovation Early Career Researcher Forum (since 2021) • Associate Fellow of AdvanceHE, UK-based society fostering higher-education quality (since 2021)

Other UK-RAS Network representative for the University of Oxford • EPSRC Peer Review College

Invited Talks and Public Engagement

2026 Presentations at Oxford Summer Schools on "How Robots Know Where They Are – Localisation and SLAM"

2026 Pint of Science @ OxTechWeek

2026 Short appearance in SBS Korean TV show *Gifted Endeavor* (youtu.be/7m19CdBHrul)

2026 Invited talk at Politecnico di Torino – Robotics Inversion: Leveraging Cloud and IoT for Control of Heterogeneous Robot Fleets at Ultra-Low Cost

2025 Invited talk at Zhejiang University

2025 Lecture at the Keble Alumni Reunion – Do Robots Really Need to Be Embodied?

2025 Invited talk at the Polytechnic of Montreal – Robotics Inversion: Leveraging Cloud and IoT for Control of Heterogeneous Robot Fleets at Ultra-Low Cost

2025 Organiser and panellist at the industrial panel "Future Telecommunication Infrastructure and Robotics for Safety-Critical Industries and Services: Industry Demands, Safety Requirements, Technical Challenges & Solution, and Economics" at the IEEE International Conference on Communications

2025 Organiser and presenter at the tutorial "Emerging Technologies, 5G and Beyond: Task-oriented Co-design of Sensing, Communication, and Control for Cyber-Physical Systems" at the IEEE Vehicular Technology Conference

2025 Invited talk at the event *Harnessing Embodied Intelligence for Defence & Security in Hazardous Environments*

2024 Invited talk at the University of Freiburg (remote) – FMCW Radar, from Off-Road to Doppler

2024 IEEE UK and Ireland Robotics and Automation Society Webinar – FMCW Radar, What's Next? From Off-Road to Doppler (tinyurl.com/3rnuk9r5)

2024 Invited to talk at the workshop *Radar in Robotics* @ ICRA (tinyurl.com/35y5xk64)

2022 Exhibitor at the University of Oxford stand at the *Amazon AWS summit* (go.aws/38Ar32t)

2021 Featured in *Autonomous Zone* on *Landrover Owner International*

2021 Presenter at the FDL 2021 Space Science & AI Showcase (bit.ly/3DbjkIN)

2021 Presenter at the SETILive event (bit.ly/3Ao3j9M)

2021 Exibitor at the ORI stand at the *Goodwood: festival of speed* (bit.ly/3fC6YZG)

2021 Featured in *Robot "training" on famous Highland estate will make driverless vehicles more reliable* (bit.ly/3zEUwl9)

2020 Presenter at the FDL 2020 Space Science & AI Showcase (bit.ly/3sgTYha)

2019 Invited talk *Condition-independent perception and navigation of complex urban environments with vision and radar*, *Robust and Generalisable Vehicle Automation* workshop @ ITSC 2019

2019 Invited talk *Overview of Deep Learning based Perception and Navigation at Oxford Robotics Institute*, *Deep learning for Automated Driving: Beyond Perception* workshop @ ITSC 2019

2019 Presenter in an internal event for members of the UK Department for Business, Energy and Industrial Strategy (BEIS)

Mentoring

Here is a list of students and PDRAs I have mentored. The publications column refers to the publications below.

		Name	Primary Advisor	Publications	Dates	After graduation
University of Oxford	PDRA	Benjamin Ramtoula	–	C46 C47 O6 O7 O9	2025 –	–
		Luke Robinson	–	O6	2026 – 2027	–
	PhD	Lianfeng Shi	D. De Martini		2026 –	–
		Kyuhwan Yeon	D. De Martini	C46 O9	2025 –	–
		Siqi Tan	D. De Martini	W22	2025 –	–
		Sitian Shen[1]	D. De Martini	O1	2024 –	–
		Halil Kelebek	D. De Martini	O5	2024 –	–
		Divya Thuremella	L. Kunze	J21 C43	2024 –	–
		Jianhao Yuan[2]	D. De Martini	J19 C45 O7	2023 –	AMI Labs
		Luke Robinson[3]	D. De Martini	C29 C32 W16 J23 J25 C47	2021 – 2026	PDRA @ Oxford
		Benjamin Ramtoula	P. Newman	C26 C30 C35 J24	2021 – 2025	PDRA @ Oxford
		Efimia Panagiotaki	D. De Martini	W17 W11 C33 C34 C38 W14 J21 C41 O2	2021 – 2026	Oxa
		Georgi Pramatarov	P. Newman	C24 C36 O10	2020 –	–
		Valentina Musat	P. Newman	C23 J15 C39	2020 – 2025	Oxa
		David Williams	P. Newman	C18 C30 C37 J16	2019 – 2024	GSK
		Luigi Campanaro	I. Havoutis	C20 C27	2018 – 2023	PDRA @ OpenUniversity
		Tim Tang[4]	P. Newman	C15 C19 J4 J9 J14 C28	2018 – 2023	Stack AV
		Roberto Aldera	P. Newman	C6 C7 J10	2017 – 2022	Thinkst Applied Research
	MPhil/Sc	Lukas Vierling	R. Clark	O6 O7	2026	–
		Samuel Sze	L. Kunze	J21 C42	2024 – 2025	A*STAR
	Visiting	François Pomerlau (Prof)	–	W20 W21 O8	2026	–
		Monica Li (PhD)[5]	PY. Lajoie		2026	–
		Effie Daum (PhD)	F. Pomerlau	W20 O8	2026	–
		Jean-Michel Fortin (PhD)	F. Pomerlau	W21	2026	–
		Giorgia Modi (MSc)	G. Averta	W18 W19	2025 – 2026	Generative Bionics
		Henry Lema (MSc)	A. Valada		2025	–
		Santosh Parit (MSc)	A. Valada	O10	2025	–
		Felix de Trogoff (MSc)	E. Frazzoli	O3	2025	French Navy
		Julia Hindel (ELLIS PhD)	A. Valada	O5 O9 O10	2024 – 2025	Oxa
		Luigi Campanaro (PDRA)	A. Bennaceur		2024 – 2025	PDRA @ EPFL
		Juan José Cabrera Mora (PhD)	A. Aparicio		2024	–
		Pierre-Yves Lajole (PhD)[6]	G. Beltrame	J24 C44	2024	AP @ Montreal
		Davide Buoso (MSc)[7]	P. Torr	J23	2024	PhD @ Turin
	CDT	Benjamin Ramtoula	P. Newman		2021	PhD @ Oxford
		Dominik Kloepfer	P. Newman		2021	PhD @ Oxford
		Prannay Kaul	P. Newman	C13	2019	PhD @ Oxford
	4Y	Thomas August	D. De Martini		2026	–
		Azka Adziman	P. Maiolino		2026	–
		Rickie Chan	D. De Martini		2026	–
		Zhongyu Shi	D. De Martini		2026	–
		Abdullah Abid	A. Trichili		2026	–
		Jaihkun Kim	D. De Martini		2025	Samsung
		Jonathan Soepadmo	D. De Martini		2025	–
		Yiming Yang	P. Newman		2024	–
		Manshika Charvi Bissessur	D. De Martini	W17	2024	PhD @ Oxford

[1] Internship @ NVIDIA
[2] Internship @ Amazon and Meta
[5] Mitacs Globalink Research Scholarship
[6] Mitacs Globalink Research Scholarship
[7] Best MSc thesis CVPL award

		Name	Primary Advisor	Publications	Dates	After graduation
		Jincheng You	D. De Martini		2024	–
		Jie Li	D. De Martini		2024	–
		Cecilia Villalpando	D. De Martini		2024	Archangel Autonomy
		Arnav Tyagi	O. Parker-Jones		2024	–
		Daoxin Zhong	D. De Martini	W16	2023	A*STAR PhD
		Fraser Rennie	D. De Martini	C30	2022	Amazon UK
		Yang Hou	P. Newman		2022	–
		Pawit Kochakarn	L. Kunze	C25	2021	Ocado UK
		Matthew Widojo	P. Newman	J18	2021	Oxa
		Chris Parsons	P. Maiolino	J11	2021	Cambridge Consultants
		Jin Yeob Chung	P. Newman		2020	–
		Zhaoyu Wang	P. Newman		2019	–
		Michael Broome	P. Newman	J8	2019	Hawk-Eye Innovations
		David Williams	P. Newman	W1 C14	2018	PhD @ Oxford
	Interns	Elise Cohen[8]	D. De Martini		2026	–
		Azka Adziman[9]	P. Maiolino		2026	–
		Rickie Chan[10]	D. De Martini		2026	–
		Jaihkun Kim[11]	D. De Martini		2025	–
		Matthew Widojo	P. Newman		2020	OXA
		Alex Goldie	P. Newman		2021	CDT @ Oxford
		Thibaut Hurson	L. Kunze		2021	–
Università degli Studi di Pavia	MSc	Filippo Cipolla	T. Facchinetti		2018	Zucchetti
		Andrea Bonandin	T. Facchinetti	C16	2018	Ariadne
		Federica Amato	T. Facchinetti		2018	Ariadne
		Andrea Torlaschi	T. Facchinetti		2017	–
		Simone Colucci	T. Facchinetti		2017	XTream
		Davide Giuffrida	T. Facchinetti		2017	PhD @ UniPV
		Giuseppe Gramazio	T. Facchinetti	J2	2017	Salesforce
		Alessandro Bertini	T. Facchinetti	C1 J2	2016	IrisCube
	BSc	Yatin Bhutani	T. Facchinetti		2017	MSc @ UniPV
		Daniele Speciale	C. Rottenbacher		2016	–
		Stefano Azzalini	C. Rottenbacher		2016	–
		Andrea Bonandin	T. Facchinetti	C3	2016	MSc @ UniPV
		Filippo Cipolla	T. Facchinetti	C2	2016	MSc @ UniPV
		Emilio Marzorati	C. Rottenbacher		2015	-
		Luca Strobino	C. Rottenbacher		2015	MSc @ PoliMI

[8] EPSRC Vacation scholarship
[9] EPSRC Vacation scholarship
[10] EPSRC Vacation scholarship
[11] EPSRC Vacation scholarship

Publication List

* in the author list stands for equal contribution.

Peer-reviewed journal articles:

J27 P. Yu, S. Saeedi, **D. De Martini**, M. Ghassemian, PJ. Sequeira Gonçalves, H. Li. Building Standards for Quantum Robotics: A Workshop at the European Robotics Forum 2027 in Birmingham [Standards]. IEEE Robotics & Automation Magazine (RAM), 2026.

J26 M. Qi, M. Gadd, **D. De Martini**, KJ. Davis, B. Xiong, A. Rosen, N. Hawes and R. Salguero-Gómez. Biodiversity research requires more motors in air, water and on land. Methods in Ecology and Evolution, 2025.

J25 L. Robinson, M. Gadd, P. Newman and **D. De Martini**. Robot-Relay: Building-Wide, Calibration-Less Visual Servoing with Learned Sensor Handover Networks. Autonomous Robots (AuRo), 2025.

J24 PY. Lajoie, B. Ramtoula, **D. De Martini**, G. Beltrame. 3D Foundation Model-Based Loop Closing for Decentralized Collaborative SLAM. *IEEE Robotics and Automation Letters*, 2025. Presented at the *International Conference on Robotics and Automation (ICRA)*, 2026.

J23 D. Buoso, L. Robinson, G. Averta, P. Torr, T. Franzmeyer* and **D. De Martini***. Select2Plan: Training-Free ICL-Based Planning through VQA and Memory Retrieval. *IEEE Robotics and Automation Letters*, 2025. Presented at the *International Conference on Robotics and Automation (ICRA)*, 2026.

J22 M. Staniaszek, T. Flatscher, J. Rowell, H. Niu, W. Liu, Y. You, M. Gadd, M. Mattamala, A. Schutz, **D. De Martini**, L. Pitt, R. Skilton, M. Fallon, N. Hawes. AutoInspect: Towards Long-Term Autonomous Inspection and Monitoring. *IEEE Transactions on Field Robotics*, 2025.

J21 E. Panagiotaki, D. Thuremella, J. Baghabrah, S. Sze, L. F. Tarimo Fu, B. Hardin, T. Reinmund, T. Flatscher, D. Marques, C. Prahacs, L. Kunze, **D. De Martini**. The Oxford RobotCycle Project: A Multimodal Urban Cycling Dataset for Assessing the Safety of Vulnerable Road Users. *IEEE Transactions on Field Robotics*, 2025. Presented at the *UK AI Research Symposium*, 2025, and at the *International Conference on Intelligent Robots and Systems (IROS)*, 2025.

J20 Y. Diao, Y. Zhang, **D. De Martini**, P. Zhao, E. Li. Task-Oriented Co-Design of Communication, Computing, and Control for Edge-Enabled Industrial Cyber-Physical Systems. *IEEE Journal on Selected Areas in Communications*, 2025.

J19 S. Agarwal, J. Yuan, P.I Newman, **D. De Martini**, M. Gadd. Bayesian Radar Cosplace: Directly Estimating Location Uncertainty in Radar Place Recognition. *in IET Radar, Sonar & Navigation*, 2025.

J18 M. Gadd, **D. De Martini**, O. Bartlett, P. Murcutt, M. Towlson, M. Widojo, V. Mușat, L. Robinson, E. Panagiotaki, G. Pramatarov, M. Kühn, L. Marchegiani, P. Newman, and L. Kunze. OORD: The Oxford Offroad Radar Dataset. *Intelligent Transportation Systems Transactions*, 2024. Presented at the *International Conference on Robotics and Automation (ICRA)*, 2024.

J17 **D. De Martini**, G. Benetti and T. Facchinetti. Real-time cyber/physical interplay in scheduling for peak load optimization in Cyber-Physical Energy Systems. *Intelligent Systems with Applications*, 2024.

J16 D. Williams, **D. De Martini**, M. Gadd and P. Newman. Mitigating Distributional Shift in Semantic Segmentation via Uncertainty Estimation from Unlabelled Data. *Transaction on Robotics*, 2024. Presented at the *40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, 2024.

J15 V. Mușat, **D. De Martini***, M. Gadd* and P. Newman. NeuralFloors: Conditional street-level scene generation from BEV semantic maps via Neural Fields. *IEEE Robotics and Automation Letters*, 2023. Presented at the *40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, 2024.

J14 T. Y. Tang, **D. De Martini**, and P. Newman. Point-based Metric and Topological Localisation between Lidar and Overhead Imagery. *Autonomous Robots (AURO)*, 2023.

J13 Z. Meng, C. She, G. Zhao, and **D. De Martini**. Sampling, Communication, and Prediction Co-Design for Synchronizing the Real-World Device and Digital Model in Metaverse. *IEEE Journal on Selected Areas in Communications*, 2023.

J12 V. Růžička, A. Vaughan, **D. De Martini**, J. Fulton, V. Salvatelli, C. Bridges, G. Mateo-Garcia, and V. Zantedeschi. RaVÆn: unsupervised change detection of extreme events using ML on-board satellites. *Nature Scientific Reports*, 2022.

J11 C. Parsons, A. Albini, **D. De Martini** and P. Maiolino. Visuo-Tactile Recognition of Partial Point Clouds Using PointNet and Curriculum Learning: Enabling Tactile Perception from Visual Data. *IEEE Robotics & Automation Magazine*, 2022. Presented at the *International Conference on Robotics and Automation (ICRA)*, 2023.

J10 R. Aldera, M. Gadd*, **D. De Martini*** and P. Newman. What Goes Around: Leveraging a Constant-curvature Motion Constraint in Radar Odometry. *IEEE Robotics and Automation Letters, IEEE*, 2022.

J9 T. Y. Tang, **D. De Martini**, S. Wu and P. Newman. Self-Supervised Learning for Using Overhead Imagery as Maps in Large-Scale Range Sensor Localisation. *International Journal of Robotics Research (IJRR)*, 2021.

J8 M. Broome, M. Gadd, **D. De Martini** and P. Newman. On the Road: Route Proposal from Radar Self-Supervised by Fuzzy LiDAR Traversability. *AI, MDPI*, 2020.

J7 **D. De Martini**, M. Gadd and P. Newman. kRadar++: Coarse-to-fine FMCW Scanning Radar Localisation. *Sensors, MDPI*, 2020.

J6 M. Musci, **D. De Martini**, N. Blago, T. Facchinetti and M. Piastra. Online Fall Detection using Recurrent Neural Networks on Smart Wearable Devices. *IEEE Trans. Emerging Topics in Computing*, 2020.

J5 **D. De Martini** and T. Facchinetti. Fault Detection of Electromechanical Actuators via Automatic Generation of Fuzzy Systems. *IEEE/ASME Trans. Mechatronics*, 2020.

J4 T. Y. Tang, **D. De Martini**, D. Barnes and P. Newman. RSL-Net: Localising in Satellite Images From a Radar on the Ground. *IEEE Robotics and Automation Letters, IEEE*, 2020.

J3 **D. De Martini**, G. Benetti, M. L. Della Vedova and T. Facchinetti. Adaptive Real-Time Scheduling of Cyber-Physical Energy Systems. *ACM Transactions on Cyber-Physical Systems, ACM*, 2017.

J2 **D. De Martini**, G. V. Gramazio, A. Bertini, C. E. Rottenbacher and T. Facchinetti. Design and Modeling of a Quadcopter with Double Axis Tilting Rotors. *Unmanned Systems, World Scientific*, 2017.

J1 M. A. Koledoye, **D. De Martini**, M. Carvani and T. Facchinetti. Design of a Mobile Robot for Air Ducts Exploration. *Robotics, MDPI*, 2017.

Peer-reviewed conference papers:

C47 L. Robinson, B. Ramtoula, A. Izaaryene, P. Newman and **D. De Martini**. Multi-Robot Planning and Control from CCTV Camera Networks in a Real Warehouse. *International Symposium of Robotics Research (ISRR)*, 2026.

C46 K. Yeon, B. Ramtoula and **D. De Martini**. PriorEye: Geospatial Visual Priors for End-to-End Autonomous Driving. Spotlight presentation at the *European Conference on Computer Vision (ECCV)*, 2026.

C45 J. Yuan, F. Pizzati, F. Pinto, L. Kunze, I. Laptev, P. Newman, P. Torr and **D. De Martini**. LikePhys: Evaluating Intuitive Physics Understanding in Video Diffusion Models via Likelihood Preference. *International Conference on Learning Representations (ICLR)*, 2026

C44 B. Ramtoula, PY. Lajoie, P. Newman and **D. De Martini**. Fantastic Features and Where to Find Them: A Probing Method to Combine Features from Multiple Foundation Models. *Conference on Neural Information Processing Systems (NeurIPS)*, 2025.

C43 D. Thuremella, Y. Yang, S. Wanna, L. Kunze and **D. De Martini**. Ensemble of Pre-Trained Models for Long-Tailed Trajectory Prediction. *IEEE Intelligent Transportation Systems Conference (ITSC)*, 2025.

C42 S. Sze, **D. De Martini** and L. Kunze. MinkOcc: Towards real-time label-efficient semantic occupancy prediction. *International Conference on Intelligent Robots and Systems (IROS)*, 2025.

C41 E. Panagiotaki, G. Pramatarov, L. Kunze and **D. De Martini**. GraphSCENE: On-Demand Critical Scenario Generation for Autonomous Vehicles in Simulation. *International Conference on Intelligent Robots and Systems (IROS)*, 2025. 🏆[12].

C40 G. Caroleo, A. Albini, **D. De Martini**, T. D. Barfoot and P. Maiolino. Tiny Lidars for Manipulator Self-Awareness: Sensor Characterization and Initial Localization Experiments. *International Conference on Intelligent Robots and Systems (IROS)*, 2025.

C39 V. Mușat, **D. De Martini***, M. Gadd* and P. Newman. NeuralFloors++: Consistent Street-Level Scene Generation From BEV Semantic Maps. *International Conference on Intelligent Robots and Systems (IROS)*, 2024. Also presented at the *UK AI Research Symposium*, 2025.

C38 E. Panagiotaki, T. Reinmund, S. Sze, B. Liu, S. Mouton, L. Pitt, A. S. Shanthini, M. Towlson, W. Tubby, C. Prahacs, **D. De Martini**, L. Kunze. RobotCycle: Assessing Cycling Safety in Urban Environments. Workshop on Socially Interactive Autonomous Mobility at the *IEEE Intelligent Vehicles Symposium*, 2024.

C37 D. Williams, M. Gadd, P. Newman and **D. De Martini**. Masked γ-SSL: Learning Uncertainty Estimation via Masked Image Modeling. *International Conference on Robotics and Automation (ICRA)*, 2024.

C36 G. Pramatarov, M. Gadd, P. Newman and **D. De Martini**. That's My Point: Compact Object-centric LiDAR Pose Estimation for Large-scale Outdoor Localisation. *International Conference on Robotics and Automation (ICRA)*, 2024.

C35 B. Ramtoula, **D. De Martini***, M. Gadd* and P. Newman. *VDNA-PR*: Using General Dataset Representations for Robust Sequential Visual Place Recognition. *International Conference on Robotics and Automation (ICRA)*, 2024.

[12] Finalist of the IEEE-RAS Outstanding Women in Robotics & Automation (WiRA) Paper Awards @ IROS25

C34 E. Panagiotaki, **D. De Martini**, and L. Kunze. Semantic Interpretation and Validation of Graph Attention-based Explanations for GNN Pose Estimation Models *International Conference on Advanced Robotics (ICAR)*, 2023.

C33 E. Panagiotaki, **D. De Martini**, G. Pramatarov, M. Gadd and L. Kunze. SEM-GAT: Lightweight, Introspective, Semantic Pose Estimation with Learned Graph Attention. *International Conference on Advanced Robotics (ICAR)*, 2023.

C32 L. Robinson, M. Gadd, P. Newman and **D. De Martini**. Robot-Relay: Building-wide, Calibration-less Visual Servoing with Learned Sensor Handover Networks. *International Symposium on Experimental Robotics (ISER)*, 2023.

C31 M. Gadd, B. Ramtoula, **D. De Martini** and P. Newman. What you see is what you get: Experience ranking with deep neural dataset-to-dataset similarity for topological localisation. *International Symposium on Experimental Robotics (ISER)*, 2023.

C30 F. Rennie, D. Williams, P. Newman and **D. De Martini**. Doppler-aware Odometry from FMCW Scanning Radar. *IEEE Intelligent Transportation Systems Conference (ITSC)*, 2023.

C29 L. Robinson, **D. De Martini**, M. Gadd and P. Newman. Visual Servoing on Wheels: Robust Robot Orientation Estimation in Remote Viewpoint Control. *International Conference on Intelligent Robots and Systems (IROS)*, 2023.

C28 T. Y. Tang, **D. De Martini** and P. Newman. Self-Supervised Lidar Place Recognition in Overhead Imagery Using Unpaired Data. *Robotics: Science and Systems (RSS)*, 2023.

C27 L. Campanaro, **D. De Martini**, S. Gangapurwala, W. Merkt and I. Havoutis. Roll-Drop: Accounting for Observation Noise with a Single Parameter. *Learning for Dynamics & Control Conference (L4DC)*, 2023.

C26 B. Ramtoula, M. Gadd, P. Newman and **D. De Martini**. Visual DNA: Representing and Comparing Images using Distributions of Neuron Activations. *Computer Vision and Pattern Recognition Conference (CVPR)*, 2023. Also presented at the *UK AI Research Symposium*, 2025, and at the *DataCV Workshop* at the *International Conference on Computer Vision (ICCV)*, 2025. 🏆[13]

C25 P. Kochakarn, **D. De Martini**, D. Omeiza, and L. Kunze. Explainable Action Prediction through Self-Supervision on Scene Graphs. *International Conference on Robotics and Automation (ICRA)*, 2023.

C24 G. Pramatarov, **D. De Martini***, M. Gadd* and P. Newman. BoxGraph: Semantic Place Recognition and Pose Estimation from 3D LiDAR. *International Conference on Intelligent Robots and Systems (IROS)*, 2022

C23 V. Mușat, **D. De Martini***, M. Gadd* and P. Newman. Depth-SIMS: Semi-Parametric Image and Depth Synthesis. *International Conference on Robotics and Automation (ICRA)*, 2022.

C22 R. Weston, M. Gadd, **D. De Martini** and P. Newman. Fast-MbyM: Leveraging Translational Invariance of the Fourier Transform for Efficient and Accurate Radar Odometry. *International Conference on Robotics and Automation (ICRA)*, 2022.

C21 M. Gadd, **D. De Martini**, and P. Newman. Contrastive Learning for Unsupervised Radar Place Recognition. *International Conference on Advanced Robotics (ICAR)*, 2021.

C20 L. Campanaro, S. Gangapurwala, **D. De Martini**, W. Merkt and I. Havoutis. Reinforcement Learning for Central Pattern Generators. *Towards Autonomous Robotic Systems Conference (TAROS)*, 2021.

C19 T. Y. Tang, **D. De Martini** and P. Newman. Get to the Point: Learning Lidar Place Recognition and Metric Localisation Using Overhead Imagery. *Robotics: Science and Systems (RSS)*, 2021.

C18 D. Williams, **D. De Martini**, M. Gadd, and P. Newman. Fool Me Once: Robust Selective Segmentation via Out-of-Distribution Detection with Contrastive Learning. *International Conference on Robotics and Automation (ICRA)*, 2021.

C17 C. Schroeder de Witt, C. Tong, V. Zantedeschi, **D. De Martini**, F. Kalaitzis, M. Chantry, P. Bilinski and D. Watson-Parris. RainBench: Towards Data-Driven Global Precipitation Forecasting from Satellite Imagery. *AAAI Conference on Artificial Intelligence*, 2021.

C16 T. Facchinetti, A. Bonandin, G. Benetti and **D. De Martini**. Distributed architecture for a smart LEDs display system based on MQTT. *IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, 2020.

C15 T. Y. Tang, **D. De Martini**, S. Wu and P. Newman. Self-Supervised Localisation between Range Sensors and Overhead Imagery. *Robotics: Science and Systems (RSS)*, 2020.

C14 D. Williams, **D. De Martini**, M. Gadd, L. Marchegiani and P. Newman. Keep off the Grass: Permissible Driving Routes from Radar with Weak Audio Supervision. *IEEE Intelligent Transportation Systems Conference (ITSC)*, 2020.

[13] Best Paper Award @ DataCV workshop @ ICCV 2025

C13 P. Kaul, **D. De Martini**, M. Gadd and P Newman. RSS-Net: weakly-supervised multi-class semantic segmentation with FMCW radar. *IEEE Intelligent Vehicles Symposium (IV)*, 2020.

C12 M. Gadd, **D. De Martini** and P. Newman. Look Around You: Sequence-based Radar Place Recognition with Learned Rotational Invariance. *IEEE/ION Position, Location and Navigation Symposium (PLANS)*, 2020.

C11 Ș. Săftescu, M. Gadd, **D. De Martini**, D. Barnes and P. Newman. Kidnapped Radar: Topological Radar Localisation using Rotationally-Invariant Metric Learning. *International Conference on Robotics and Automation (ICRA)*, 2020.

C10 T. Tse, **D. De Martini** and L. Marchegiani. No Need to Scream: Robust Sound-based Speaker Localisation in Challenging Scenarios *International Conference on Social Robotics*, 2019.

C9 D. Giuffrida, G. Benetti, **D. De Martini** and T. Facchinetti. Fall Detection with Supervised Machine Learning using Wearable Sensors *International Conference on Industrial Informatics*, 2019.

C8 S. Kyberd, J. Attias, P. Murcutt, C. Prahacs, M. Towlson, S. Venn, A. Vasconcelos, M. Gadd, **D. De Martini** and P. Newman. The hulk: design and development of a weather-proof vehicle for long-term autonomy in outdoor environments. *International Conference on Field and Service Robotics*, 2019.

C7 R. Aldera, **D. De Martini***, M. Gadd* and P. Newman. What Could Go Wrong? Introspective Radar Odometry in Challenging Environments. *IEEE Intelligent Transportation Systems Conference (ITSC)*, 2019.

C6 R. Aldera, **D. De Martini***, M. Gadd* and P. Newman. Fast radar motion estimation with a learnt focus of attention using weak supervision. *IEEE International Conference on Robotics and Automation (ICRA)*, 2019.

C5 M. Koledoye, **D. De Martini**, S. Rigoni and T. Facchinetti. A Comparison of RSSI Filtering Techniques for Range-based Localization. *IEEE International Conference on Emerging Technologies and Factory Automation*, 2018.

C4 **D. De Martini**, G. Benetti and T. Facchinetti. Cyber/physical interplay in the real-time scheduling for peak load optimization of electric loads. *IEEE International Conference on Industrial Cyber-Physical Systems*, 2018.

C3 **D. De Martini**, A. Bonandin and T. Facchinetti. eduMorse: An Open-Source Framework for Mobile Robotics Education. *Robotics in Education: Latest Results and Developments*, 2017.

C2 **D. De Martini**, G. Benetti, F. Cipolla, D. Caprino, M. L. Della Vedova and T. Facchinetti. Peak load optimization through 2-dimensional packing and multi-processor real-time scheduling. *Proceedings of ACM International Conference on Computing Frontiers*, 2017. 🏆[14]

C1 **D. De Martini**, G. Roveda, A. Bertini, A. Marchini and T. Facchinetti. A Framework for Automatic Generation of Fuzzy Evaluation Systems for Embedded Applications. *International Joint Conference on Computational Intelligence*, 2017.

Peer-reviewed workshop papers:

W22 S. Tan, D. He, Y. Zhang, **D. De Martini**, and M. Fallon. Training-Free Open-Vocabulary Dense Mapping via Exchanged Semantic–Geometric Verifications. Workshop on *Semantic-Aware Mapping and Navigation* in the *International Conference on Intelligent Robots and Systems (IROS)*, 2026.

W21 JM. Fortin, **D. De Martini**, F. Pomerleau and P. Giguère. Inferring Terrain Stability from Proprioceptive Data of a Quadruped Robot. Workshop on *Field Robotics* in the *International Conference on Robotics and Automation (ICRA)*, 2026.

W20 E. Daum, **D. De Martini** and F. Pomerleau. A Novel Trajectory Evaluation Metric: Distance Travelled Error (DTE). Workshop on *Field Robotics* in the *International Conference on Robotics and Automation (ICRA)*, 2026.

W19 G. Modi, D. Buoso, G. Averta, **D. De Martini**. RGB-only Active 3D Scene Graph Generation for Indoor Mobile Robots. Workshop on *Semantics for Reliable Robot Autonomy* in the *International Conference on Robotics and Automation (ICRA)*, 2026.

W18 G. Modi, D. Buoso, G. Averta, **D. De Martini**. Fixed External Cameras as Common Prior Maps for Active 3D Scene Graph Generation. Workshop on *Robots Meet Prior Maps* in the *International Conference on Robotics and Automation (ICRA)*, 2026.

W17 M. Bissessur, E. Panagiotaki, **D. De Martini**. Introspection in Learned Semantic Scene Graph Localisation. Workshop on *Fully Autonomy Emerges from Cognition (FAST)* in the *International Conference on Intelligent Robots and Systems (IROS)*, 2025.

W16 D. Zhong, L. Robinson and **D. De Martini**. NeRFoot: Robot-Footprint Estimation for Image-Based Visual Servoing. ICRA40 anniversary, 2024.

[14] Winner of the Best Short Paper award.

W15 M. Gadd, **D. De Martini**, O. Bartlett, P. Murcutt, M. Towlson, M. Widojo, V. Musat, L. Robinson, E. Panagiotaki, G. Pramatarov, M. Alexander Kühn, L. Marchegiani, P. Newman and L. Kunze. OORD: The Oxford Offroad Radar Dataset. Workshop on Resilient Off-road Autonomy at the *International Conference on Robotics and Automation (ICRA)*, 2024.

W14 E. Panagiotaki, T. Reinmund, S. Sze, B. Liu, S. Mouton, L. Pitt, A. S. Shanthini, M. Towlson, W. Tubby, C. Prahacs, **D. De Martini**, L. Kunze. RobotCycle: Assessing Cycling Safety in Urban Environments. Workshop on Field Robotics at the *International Conference on Robotics and Automation (ICRA)*, 2024.

W13 M. Gadd, **D. De Martini**, L. Pitt, W. Tubby, M. Towlson, C. Prahacs, O. Bartlett, J. Jackson, M. Qi, P. Newman, A. Hector, R. Salguero-Gómez, N. Hawes. Watching Grass Grow: Long-term Visual Navigation and Mission Planning for Autonomous Biodiversity Monitoring. Workshop on Field Robotics at the *International Conference on Robotics and Automation (ICRA)*, 2024.

W12 Y. Diao, H. Dai, G. Zhao, and **D. De Martini**. TAGIC: Task-Guided Image Communication Framework for Seamless Teleoperation. Poster at *IEEE International Conference on Computer Communications (INFOCOM)*, 2024.

W11 E. Panagiotaki, **D. De Martini**, and L. Kunze. Towards Semantic Interpretation and Validation of Graph Attention-based Explanations. Explainable Robotics Workshop at the *International Conference on Robotics and Automation (ICRA)*, 2023.

W10 Z. Meng, C. She, G. Zhao, **D. De Martini**, and Y. Zhang. Sampling, Communication, and Prediction Co-Design for Synchronizing the Real-Robot and Digital-Robot in Metaverse. Late Breaking Results Poster at the *International Conference on Intelligent Robots and Systems (IROS)*, 2022

W9 Y. Diao, H. Dai, G. Zhao, and **D. De Martini**. Keyframe Selection, Communication, and Prediction for Teleoperated Driving Systems. Late Breaking Results Poster at the *International Conference on Intelligent Robots and Systems (IROS)*, 2022

W8 V. Růžička, A. Vaughan, **D. De Martini**, J. Fulton, V. Salvatelli, C. Bridges, G. Mateo-Garcia, and V. Zantedeschi. Smart satellites: machine learning on-board for low-latency novelty detection at the *AGU Fall Meeting*, 2021.

W7 V. Růžička, A. Vaughan, **D. De Martini**, J. Fulton, V. Salvatelli, C. Bridges, G. Mateo-Garcia, and V. Zantedeschi. Unsupervised Change Detection of Extreme Events Using ML On-Board. Workshop on Artificial Intelligence for Humanitarian Assistance and Disaster Response Workshop (AI+HADR) at the *Conference on Neural Information Processing Systems (NeurIPS)*, 2021.

W6 T. Suleymanov, M. Gadd, **D. De Martini**, and P. Newman. The Oxford Road Boundaries Dataset. Workshop on 3D-Deep Learning for Automated Driving at the *IEEE Intelligent Vehicles Symposium (IV)*, 2021.

W5 M. Gadd*, **D. De Martini*** and P. Newman. Unsupervised Place Recognition with Deep Embedding Learning over Radar Videos. All-Weather Autonomy Workshop at the *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.

W4 C. Tong, C. Schroeder de Witt, V. Zantedeschi, **D. De Martini**, F. Kalaitzis, M. Chantry, P. Bilinski and D. Watson-Parris. RainBench: Enabling Data-Driven Precipitation Forecasting on a Global Scale. Tackling Climate Change with Machine Learning and AI for Earth Sciences workshops at the *Conference on Neural Information Processing Systems (NeurIPS)*, 2020.

W3 V. Zantedeschi, **D. De Martini**, C. Tong, C. Schroeder de Witt, F. Kalaitzis, P. Biliński, M. Chantry and D. Watson-Parris. Towards Data-Driven Physics-Informed Global Precipitation Forecasting from Satellite Imagery. Tackling Climate Change with Machine Learning and AI for Earth Sciences workshops at the *Conference on Neural Information Processing Systems (NeurIPS)*, 2020.

W2 M. Gadd*, **D. De Martini***, L. Marchegiani, P. Newman and L. Kunze. Sense-Assess-eXplain (SAX): building trust in autonomous vehicles in challenging real-world driving scenarios. Workshop on Ensuring and Validating Safety for Automated Vehicles (EVSAV) at the *IEEE Intelligent Vehicles Symposium (IV)*, 2020.

W1 D. Williams, **D. De Martini**, L. Marchegiani and P. Newman. Listening closely to see far away: Radar-based Terrain Classification from Auditory Signals. Abstract submission at the *International Conference on Digital Image & Signal Processing*, 2019.

Books:

B1 C. Cusano, T. Facchinetti, C. Larizza, E. Losiouk, A. Tramonte and **D. De Martini**. Programmazione In Linguaggio C. Problemi e Temi Svolti. (eng: Exercises in the C Programming Language) *Medea Editore*, 2015.

Online:

O10 S. Parit, J. Hindel, R. Mohan, G. Pramatarov, P. Newman, **D. De Martini**, A. Valada. Real-Time Radar Semantic and Panoptic Scene Understanding. Arxiv: https://arxiv.org/abs/2610.26816, 2026.

O9 J. Hindel, B. Ramtoula, K. Yeon, N. Heppert, P. Newman, **D. De Martini**, and A. Valada. Diversity Without Labels: Task-Agnostic Coreset Selection Across Perception and Action Arxiv: https://arxiv.org/abs/2610.23642, 2026.

O8 E. Daum, **D. De Martini**, C. Dune, and F. Pomerleau. Comparing Trajectories from Positions Alone: Curvature-Based Time Alignment and Drift Error Metric. Arxiv: https://arxiv.org/abs/2609.14936, 2026

O7 J. Yuan, Y. Yuan, B. Ramtoula, L. Vierling, P. Newman, L. Kunze, P. Torr, **D. De Martini**. Beyond Visual Quality: A Study of Test-Time Planning with World Action Models. Arxiv: https://arxiv.org/abs/2609.24745, 2026

O6 L. Vierling, B. Ramtoula, L. Robinson, R. Clark, **D. De Martini**. InfraVLA: Extending Vision-Language-Action Navigation with Infrastructure Cameras. ArxivL https://arxiv.org/abs/2609.33647, 2026.

O5 H. Kelebek, J. Hindel, K. Hoffman, L. Hoffman, A. Loveridge, B. Mandinyenya, K. Ncube, J. Seymour-Smith, A. Sibanda, A. Valada, M. Wijers, and **D. De Martini**. Spotted: Location-informed Reidentification of Hyenas and Leopards in Camera Trap Surveys. Arxiv: https://arxiv.org/abs/2607.00804, 2026.

O4 B. Hardin, E. Panagiotaki, **D. De Martini** and L. Kunze. What Can Eye Gaze Teach Us About Real-World Cycling? Insights From the Oxford RobotCycle Project. Arxiv: https://arxiv.org/abs/2606.04989, 2026.

O3 F. de Trogoff du Boisguezennec, B. Ramtoula, and **D. De Martini**. Beyond a Shadow of a Doubt: Close Proximity Geometry Reconstruction Using FMCW Radar Shadow Effects. Arxiv: https://arxiv.org/abs/2606.25829, 2026.

O2 E. Panagiotaki, **D. De Martini**, L. Kunze, P. Veličković. NAR-* ICP: Neural Execution of Classical ICP-based Pointcloud Registration Algorithms. Arxiv: https://arxiv.org/pdf/2410.11031, 2025.

O1 S. Shen, G. Pramatarov, Y. Tao, **D. De Martini**. Neural-MMGS: Multi-modal Neural Gaussian Splats for Large-Scale Scene Reconstruction. Arxiv: https://arxiv.org/pdf/2509.17762, 2025.